PRODUCT REQUIREMENTS DOCUMENT (PRD)

DSU Faculty Research Intelligence, Automated Data Synchronization & Accreditation Reporting System

Detailed Version — Written so that faculty, research coordinators, administrators, developers and project evaluators can understand the complete system.

1. Executive Summary

The proposed system is a centralized faculty research information and accreditation-reporting application for Dayananda Sagar University (DSU). The system is designed to reduce repetitive manual entry of research information such as journal publications, conference papers, books, patents, funded projects and related academic activities.

A faculty member will be associated with research identifiers such as ORCID iD and, where available and authorized, Scopus Author ID. The application will use official APIs or other authorized data-import mechanisms to retrieve research metadata. The retrieved information will be matched to the correct DSU faculty member, checked for duplicates, placed into a verification workflow and then stored in the central DSU research database.

The same verified database will be used to prepare accreditation-related reports. NAAC and NBA are treated as separate reporting frameworks. The application will map the master data to the applicable current official criteria, sub-criteria, tables and evidence requirements and generate the required output for the selected accreditation framework.

2. Problem Being Solved

• Faculty/research staff may have to enter the same publication or research information multiple times.

• Publication information can change or become outdated as new papers, citations or research activities are added.

• Different departments may maintain information in different spreadsheets or documents.

• Preparing faculty-wise, department-wise and academic-year-wise research summaries is time-consuming.

• Accreditation reporting requires information in structured criteria/table formats, so manually converting a general research list into the required report is inefficient.

• NAAC and NBA have different purposes and reporting structures, so one generic report cannot safely be used for both.

3. Product Objective

Main objective: Create one trusted DSU faculty research database that can be continuously updated from authorized sources and reused to produce accurate, traceable and accreditation-specific reports.

The application should follow the principle: **Enter/identify once → verify once → reuse many times.**

4. Users and Roles

Role	Main responsibilities
System Administrator	Manage users, departments, faculty profiles, integrations, system settings and permissions.
Research / Accreditation Coordinator	Review research records, verify imported publications, manage accreditation mappings, generate reports.
Faculty	View profile, review imported research records, provide identifiers, request corrections and upload supporting evidence wh
Department Administrator / HOD	Review department research information and generate department-level reports.
Viewer / Management	View approved dashboards and reports without modifying source data.

5. Faculty Profile

Every faculty member will have a centralized profile. The profile is the link between the DSU identity and external research identities.

Field	Purpose
Faculty ID / Employee ID	Unique DSU identity.
Name	Official faculty name.
Department	Department to which the faculty belongs.
Designation / Cadre	Professor, Associate Professor, Assistant Professor, etc., as applicable.
ORCID iD	Researcher identifier used for authorized ORCID integration.
Scopus Author ID	Scopus researcher identifier when available.
Other identifiers	Optional identifiers such as institutional research IDs or other approved sources.
Academic year	Used for year-wise reporting.
Profile status	Active/inactive and verification status.

6. Research Data Modules

6.1 Journal Publications

Store title, author list, journal/source, DOI, ISSN, publication year/date, publisher, volume/issue/pages where available, indexing information such as Scopus/WoS/SCI where verified, quartile information where applicable, citations and source.

6.2 Conference Publications

Store title, authors, conference name, event date/year, national/international classification, publisher/organizer, DOI/proceedings information and supporting evidence.

6.3 Books and Book Chapters

Store title, author/editor, publisher, ISBN, publication year, chapter title where applicable and book/book-chapter classification.

6.4 Patents

Store title, inventors, application/patent number, filing date, publication date, status such as filed/published/granted where applicable, jurisdiction and supporting document.

6.5 Sponsored Research / Funding

Store project title, funding agency, PI, co-investigators, sanctioned amount, start/end dates, project status and supporting evidence.

6.6 Courses / Academic Research Activities

Store relevant course/certification/research-development activity information required by the institution or applicable accreditation template.

6.7 Citations and Metrics

Store source, metric value, date captured and provenance. Metrics should be treated as time-dependent rather than permanent values.

7. ORCID and Scopus Integration — How It Works

The system should not describe the process as unauthorized 'scraping'. The preferred implementation is API-based or another authorized data-import method. ORCID and Scopus have different roles: ORCID is a researcher identity and works repository; Scopus is an indexing/database service. The system may use ORCID to identify a researcher and, where DSU has authorized Scopus access, use the appropriate Elsevier/Scopus API to retrieve Scopus-indexed metadata.

Example user journey:

• Administrator creates the DSU faculty profile.

• Administrator enters/associates the faculty member's ORCID iD and/or Scopus Author ID.

• System validates the identifier and retrieves available information through the authorized integration.

• System attempts to match the external author to the DSU faculty profile.

• System imports available publication metadata.

• System checks whether each publication already exists.

• New or changed records are placed into Pending Verification.

• Faculty/research coordinator verifies the record.

• Verified information becomes part of the official DSU research database.

• Accreditation reports and dashboards use the verified record.

High-level flow:

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DSU Faculty → ORCID/Scopus Identifier → Authorized API → Research Metadata → Author Matching →
Duplicate Check → Verification → DSU Master Database → Accreditation Mapping → NAAC/NBA Report
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8. Synchronization

The application should support both an initial import and repeated synchronization. A scheduled job may check for new or updated records. The exact frequency should depend on the API's terms, limits and DSU requirements.

Synchronization field	Example
Last synchronization	05-Oct-2026 18:00
Source	Scopus API / ORCID API / approved import
Records found	18
New records	3
Updated records	2
Duplicates skipped	13
Pending verification	3
Sync status	Successful / Partial / Failed

9. Verification and Data Quality

Automatic retrieval should not mean automatic acceptance as official accreditation evidence. A verification layer protects the institution against wrong author matches, duplicate records and incorrect metadata.

• Pending: imported but not yet verified.

• Verified: checked and approved for official use.

• Rejected: determined to belong to another researcher or otherwise invalid.

• Corrected: administrator/faculty corrected metadata while retaining an audit trail.

• Source and synchronization timestamp should be retained.

• DOI, title, author names and other metadata can be used for duplicate detection.

• Important changes should be auditable.

10. Master Database Design

The master database should store research information once. NAAC and NBA should not have separate copies of the same publication data.

Table / entity	Purpose
faculty	DSU faculty identity and profile.
departments	Department master data.
research_identifiers	ORCID, Scopus Author ID and other identifiers.
publications	Common publication metadata.
publication_authors	Many-to-many relationship between publications and authors/faculty.
journals	Journal/source metadata and classification information.
conferences	Conference metadata.
books	Books and chapters.
patents	Patent information.
research_projects	Projects, grants and funding.
evidence_documents	Supporting files or evidence references.
sync_logs	Integration history.
verification_records	Approval/rejection/correction history.
accreditation_templates	Versioned NAAC/NBA template definitions.
criteria_mappings	Mapping from accreditation fields to database fields.
generated_reports	Report metadata and generation history.

11. NAAC and NBA — Why Separate Formats Are Required

The application must not treat NAAC and NBA as two names for the same PDF. They are different accreditation frameworks. NAAC is institution-level accreditation, while NBA accredits specific academic programs. NBA therefore requires the system to select the applicable program/manual/SAR format.

The exact criteria and tables can also change when the authorities publish revised manuals. Therefore, the software should use a versioned criteria/template engine rather than hard-coding one permanent PDF.

12. Official Criteria as the Source of Truth

Before configuring the production report templates, the project team must obtain the latest official documents applicable to DSU. For NBA, this means selecting the current official manual and SAR/evaluation documents applicable to the relevant DSU engineering program and accreditation cycle. For NAAC, it means selecting the current official framework/manual/template applicable to DSU and the relevant assessment cycle.

The system should record which official version was used to generate each report. If an authority releases a new version, a new template version can be created without destroying historical mappings.

13. Accreditation Criteria Mapping Engine

This is a central feature. The application converts official accreditation requirements into configurable software rules.

Layer	What the software stores
Framework	NAAC or NBA.
Program / scope	Institution-level NAAC or specific NBA program, e.g. applicable engineering program.
Manual version	Official document/version used.
Criterion	Criterion number/name from the official document.
Sub-criterion	Sub-criterion/parameter where applicable.
Table / item	Required table or reporting item.
Required fields	Data that must be populated.
Evidence	Supporting documents or references required/allowed.
Calculation	Any prescribed count, ratio, percentage or other derived value.
Output layout	Where the data appears in the generated report.

14. Example of Database-to-Accreditation Mapping

A faculty publication exists once in the master database. Suppose the record contains title, faculty, journal, DOI, year, indexing and other metadata. The mapping engine decides which fields are needed by the selected current NAAC or NBA table. The same publication can therefore contribute to multiple reports without being duplicated in the database.

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Conceptual flow:

Publication record → Criteria mapping → Required fields → Validation → Report table → PDF
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15. NBA Reporting

For engineering programs, the software must select the applicable current NBA UG Engineering category/manual and corresponding SAR format. The current NBA documentation contains program-specific criteria and prescribed reporting/evidence requirements, including faculty-related and research-related information. The application should therefore map faculty research data to the exact applicable criterion/table rather than assume a universal 'NBA faculty PDF'.

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Example configuration:

Accreditation = NBA Program = DSU engineering program Manual/SAR = applicable current official version
Academic Year = 2025–26 Section = selected criterion/table Output = prescribed SAR/report structure
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16. NAAC Reporting

NAAC reporting must similarly use the current official NAAC framework/template applicable to DSU. Since NAAC is institution-focused, the application may need institutional-level aggregation rather than only individual-faculty reports. The research database can provide the underlying faculty/research information required by the applicable NAAC criteria and data tables.

17. Report Generation User Interface

User selects	Possible values
Accreditation framework	NAAC / NBA
Applicable version	Current official template/manual version available to administrator
Academic year	2025–26, 2026–27, etc.
Department	CSE / ECE / etc.
Faculty	Individual / selected group / all
Report section	Applicable criterion/table/section
Output	PDF / on-screen preview / approved export

18. Validation Before PDF

Before generating the final report, the system should check required fields. Example: if an official table requires publication year and DOI/source information, the system should identify missing values rather than silently generate an incomplete report.

• Required field missing → show warning.

• Unverified record → mark clearly and optionally exclude from official report.

• Invalid academic year → prevent submission.

• Duplicate record → block or request review.

• Evidence missing where required → flag before final generation.

19. Evidence Management

Research records may require supporting evidence such as publication PDFs, acceptance letters, patent documents, project sanction letters or other institutional documents. The application should allow authorized users to attach or reference evidence and associate it with the relevant research record and accreditation criterion.

20. Dashboard and Analytics

• Total faculty research records.

• Publication count by year.

• Journal vs conference vs book distribution.

• Department-wise research output.

• Patent and funded-project counts.

• Research activity by faculty.

• Indexing/category summaries where verified.

• Pending verification count.

• Data completeness for accreditation reporting.

21. Audit Trail and Security

• Role-based access control.

• Backend-only storage of API credentials.

• Log synchronization attempts and results.

• Record who verified or modified important research data.

• Keep previous values/version history where needed.

• Restrict official report generation to authorized users.

• Protect faculty personal and institutional information.

22. Error Handling

Scenario	Expected system behaviour
API unavailable	Keep existing data, record failed sync, allow retry.
Faculty not found	Show unresolved identity for administrator review.
Multiple author matches	Do not automatically select; request verification.
Duplicate publication	Flag and avoid duplicate insertion.
Incomplete metadata	Import what is available and mark missing fields.
Rate/API limit	Respect source limits and schedule a later retry.
Template changed	Create new template version rather than altering historical reports.

23. End-to-End Example

Step 1: DSU administrator creates Dr. ABC's faculty profile.

Step 2: Administrator associates ORCID iD and, where available, Scopus Author ID.

Step 3: System retrieves available authorized research data.

Step 4: System identifies 10 new publications and 2 updated records.

Step 5: Duplicate detection removes/skips records already present.

Step 6: The 10 new records appear as Pending Verification.

Step 7: Faculty/research coordinator verifies the publications.

Step 8: Verified records are added to the official DSU research dataset.

Step 9: Coordinator selects the applicable current NBA report/template and academic year.

Step 10: System maps the verified data to the required criteria/table fields, identifies missing evidence, and generates the prescribed report output.

Step 11: The same master data can later be mapped to the applicable NAAC report without re-entering the publications.

24. Non-Goals / Important Constraints

• The system is not a replacement for Scopus, Web of Science or ORCID.

• It must not bypass authentication, subscriptions, API limits or access controls.

• It must not assume that a current-looking PDF found online is the official applicable accreditation template.

• It must not hard-code one NBA or NAAC version permanently.

• It must not treat automatically imported data as verified accreditation evidence without the required review.

25. Success Criteria

• Faculty research information can be maintained in one centralized database.

• Authorized research data can be synchronized with minimal manual entry.

• Imported records can be verified and traced to their source.

• Duplicate publications are prevented or flagged.

• Academic-year and department-wise reports can be generated.

• The system can map master data to the applicable current NAAC/NBA criteria and tables.

• Reports clearly identify the official template/manual version used.

• Users can generate the required PDF output without re-entering the same research data.

26. Final Product Architecture

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FACULTY PROFILE → ORCID / SCOPUS IDENTIFIER → AUTHORIZED DATA SOURCE → SYNCHRONIZATION → AUTHOR
MATCHING → DUPLICATE CHECK → VERIFICATION → DSU MASTER RESEARCH DATABASE → ANALYTICS → ACCREDITATION
CRITERIA MAPPING → NAAC / NBA TEMPLATE ENGINE → PDF REPORT
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27. Final Requirement for Project Team

The development team should not finalize the NAAC/NBA PDF layouts from memory or from an unofficial website. Before implementation, the team must obtain the official current documents applicable to DSU, identify every relevant criterion/sub-criterion/table/data field/evidence requirement, and enter those mappings into the system's versioned accreditation-template configuration. This makes the application maintainable when NAAC or NBA publishes a revised format.